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Improvement of the model proposed by Menegotto and Pinto for steel

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ABSTRACT

The model proposed by Menegotto and Pinto (1973) is widely used to simulate the dynamic response of steel structures and steel bars of reinforced concrete structures. In the early 1980s, Filippou et al. (1983) improved the original model by introducing isotropic hardening and highlighted a flaw in the original formulation. Specifically, these researchers observed that if partial unloading takes place at strains lower than the maximum recorded value, the reloading path provides forces that are higher than those expected. Filippou et al. deemed that such errors were not particularly significant for reinforced concrete members and did not propose any modification to the analytical formulation of the model. In the opinion of the writers, this observation is convincing when dealing with reinforced concrete members but is questionable when referring to steel members. The abovementioned flaw is eliminated in this paper from the Menegotto and Pinto model and the improvement over the original model is shown by means of nonlinear dynamic analyses on simple reinforced concrete and steel frames.

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1. Introduction

The model originally proposed by Menegotto and Pinto [1] is a uniaxial material model able to simulate the main aspects of the cyclic response of steel. It is widely used to simulate the dynamic response of steel members discretized by means of fibres and subjected to normal stresses (e.g. [2–4]) or the response of steel bars of reinforced concrete structures (e.g. [5–7]). Owing to the simplicity of its formulation, it is implemented in numerous programs intended to the evaluation of the response of structures, e.g. OpenSees [8]. The first formulation of the model, published in 1970 by Giuffrè and Pinto [9], drew its origin from a previous research study by Goldberg and Richard [10] and considered the possibility of unloading and reloading paths. The model proposed by Giuffrè and Pinto [9] was enhanced by Menegotto and Pinto [1,11] to simulate kinematic hardening. Then, in the early 1980s, Filippou et al. [12] improved the Menegotto and Pinto model through the introduction of the isotropic hardening. These researchers also noted that, if partial unloading and reloading paths took place at deformations lower than the maximum recorded values, the response of the model produced stresses higher than those corresponding to the previous loading curve. However, Filippou et al. [12] concluded that such errors were not particularly significant in reinforced concrete structures and did not propose any modification to the analytical formulation of the model. More recently, Zsarnóczay and Budaházy [13] suggested some modifications to the model proposed by Menegotto and Pinto in order to eliminate the shortcoming highlighted by Filippou et al. [12]. Specifically, Zsarnóczay and Budaházy recorded the parameters governing the response of all half-cycles and compared the stress at every load step to the stresses from previous cycles at the same strain level [13]. Then, the parameters of the curve were modified if the calculated stress exceeded the stress in the previous cycle by more than the isotropic hardening could account for [13]. Like the model proposed by Zsarnóczay and Budaházy [13], the uniaxial material model proposed by the authors intends to eliminate the abovementioned flaw from the model originally described by Menegotto and Pinto. The modified model retains the simplicity of the original formulation. Owing to this, it is characterized by a low calculus memory and computational time. Therefore, the proposed model is particularly suitable for use in structural analysis. The validity of the new formulation is demonstrated by comparison of the response of the current and new models on some reinforced concrete and steel members subjected to axial force and bending moment.

2. The original model by Menegotto and Pinto and successive modifications

The model was first published by Giuffrè and Pinto [9]. These researchers modified the relations previously proposed by

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Goldberg and Richard [10] so that they were apt to describe the response of steel to reversed cyclic loading. The uniaxial stress σ was calculated from a normalized stress σ^*, similarly to Ramberg and Osgood [14]. The relation that provided the normalized stress as a function of the normalized strain ε^* was

$$\sigma^* = \frac{\varepsilon^*}{(1+\varepsilon^{*R})^{\frac{1}{R}}} \tag{1}$$

and represented the equation of a curved transition from one straight line asymptote with slope E_0 equal to the initial Young's modulus of steel to another line asymptote with slope E_∞ equal to zero (Fig. 1). In Eq. (1), the parameter R influences the shape of the transition curve and takes account of the Bauschinger effect. To describe this effect accurately, the parameter R was considered to vary as a function of the plastic excursion ξ of the previous loading path as given by the following relation

$$R = R_0 - \frac{a_1\xi}{a_2+\xi} \tag{2}$$

where R_0 is the value of the parameter R during first loading and a_1 and a_2 are experimentally determined parameters.

For the curve of first loading, the relations which provide the normalized stress and strain are

$$\sigma^* = \frac{\sigma}{\sigma_0}; \quad \varepsilon^* = \frac{\varepsilon}{\varepsilon_0} \tag{3}$$

where σ_0 is the yield stress and ε_0 is the yield strain. For the curves subsequent to the first load reversal, these relations are

$$\sigma^* = \frac{\sigma-\sigma_r}{2\sigma_0}; \quad \varepsilon^* = \frac{\varepsilon-\varepsilon_r}{2\varepsilon_0} \tag{4}$$

where σ_r and ε_r are the stress and strain, respectively, at the last load reversal.

The model was subsequently enriched by Menegotto and Pinto [1,11] to take account of the kinematic hardening. The relation that provided the normalized stresses was

$$\sigma^* = b\varepsilon^* + \frac{(1-b)\varepsilon^*}{(1+\varepsilon^{*R})^{\frac{1}{R}}} \tag{5}$$

where b is the strain hardening ratio, i.e. the ratio of the elasticity modulus E_∞ of the second asymptote to the initial Young's modulus of steel. In that version of the model [1,11], Eq. (4) was substituted for

$$\sigma^* = \frac{\sigma-\sigma_r}{\sigma_0-\sigma_r}; \quad \varepsilon^* = \frac{\varepsilon-\varepsilon_r}{\varepsilon_0-\varepsilon_r} \tag{6}$$

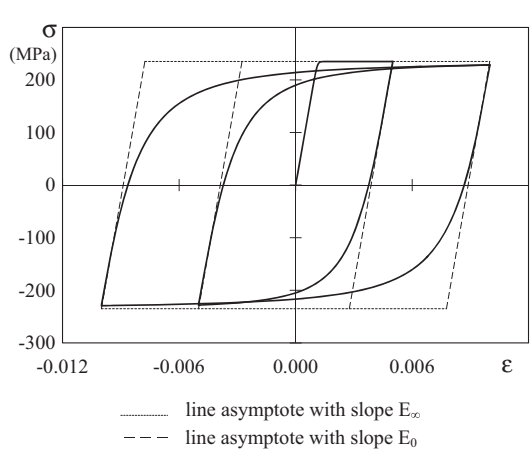

Fig. 1. Model proposed by Giuffrè and Pinto [9].

where σ_0 and ε_0 are stress and strain at the point where the two asymptotes of the branch under consideration meet and σ_r and ε_r are stress and strain at the point where the last strain reversal with stress of equal sign takes place. As an example, Fig. 2a shows the points $P_0(\varepsilon_0, \sigma_0)$ and $P_r(\varepsilon_r, \sigma_r)$ corresponding to the first load reversal of steel. The parameter R is considered dependent on the strain difference between the current asymptote intersection point and the previous load reversal point with maximum or minimum strain depending on whether the corresponding steel stress is positive or negative [12]. The parameters ε_0, σ_0, ε_r, σ_r, and R are updated after each strain reversal. As an example, Fig. 2b illustrates the change in $P_0(\varepsilon_0, \sigma_0)$ and $P_r(\varepsilon_r, \sigma_r)$ on the occurrence of four load reversals. The loading branches subsequent to the load reversals are labelled by means of roman numbers and similarly are points P_0 and P_r.

This formulation has been object of several studies intended to highlight qualities and deficiencies of the model [12,13,15–17]. Some research studies (e.g. Stanton and McNiven 1979) have showed that the explicit formulation proposed by Menegotto and Pinto [1] offers numerical efficiency and that the results provided by this model agree well with experimental results from cyclic tests on reinforcing steel bars. Some researchers (e.g. [12,13,15]) have also proposed modifications to Eq. (5) in order to improve the accuracy of the model. In particular, Filippou et al. [12] included the possibility that the model considered the isotropic hardening, as noted in laboratory tests on reinforcing steel bars. The model proposed by Filippou et al. [12] accounts for isotropic strain hardening by shifting the position of the yield asymptote before computing the new asymptote intersection point following a strain reversal. The shift is effected by moving the initial yield asymptote through a stress shift parallel to its direction. This idea was introduced some years before by Stanton and McNiven [15] but the latter researchers considered both a strain and stress shift on the monotonic envelope curve to achieve an accurate representation of the hysteretic behaviour of steel under generalized strain

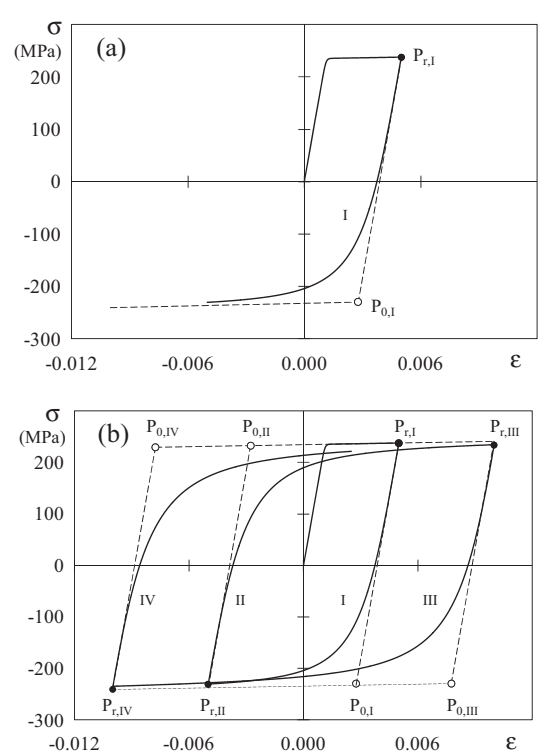

Fig. 2. Model proposed by Menegotto and Pinto [1]: (a) first cycle and (b) subsequent cycles.

histories. The yield asymptote shift suggested by Filippou et al. [12] was dependent on the maximum plastic strain, as reported in the following relation

$$\frac{\sigma_{st}}{\sigma_0} = a_3\left(\frac{\varepsilon_{max}}{\varepsilon_0} - a_4\right) \tag{7}$$

where ε_{max} is the absolute maximum strain at the instant of strain reversal and a_3 and a_4 are experimentally determined parameters.

Filippou et al. [12] also added some clarification in regard to the set of rules for unloading and reloading. In particular, Filippou et al. [12] stated that if the analytical model had a memory extending over all previous branches of the stress-strain history, it would follow the previous loading branch, as soon as the new reloading curve reached it. This would require that the model store all necessary information to retrace all previous reloading curves that were left incomplete. According to Filippou et al. [12] this was impractical from a computational standpoint. Memory of the past stress-strain history was, therefore, limited to a predefined small number of controlling curves. Due to these restrictions, reloading after partial unloading connects with curves that are different from the expected envelope curve. According to Filippou et al. [12], the discrepancy between the adopted analytical model and the actual behaviour was, however, acceptable and no remedy was offered to this shortcoming. In the opinion of the writers this statement may be questionable. To explain this misgiving, some cyclic strain histories are devised which consider some reloading branches after partial unloading. The response of the model proposed by Menegotto and Pinto [1] to these strain histories is shown in Fig. 3

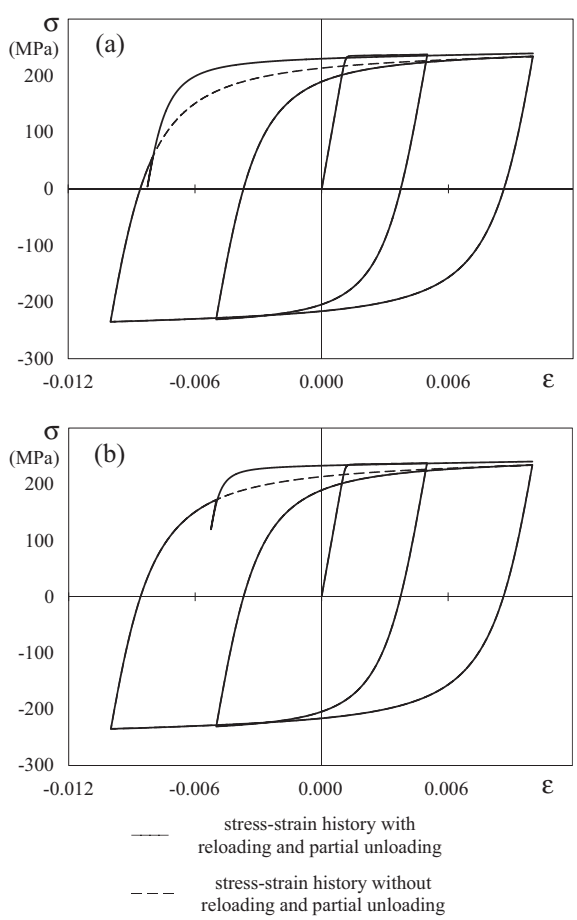

Fig. 3. Error made by the model proposed by Menegotto and Pinto [1] in case of reloading after partial unloading.

together with the response to strain histories that do not consider partial unloading branches. The differences between the two responses are evident and not negligible in the light of current scientific expectations and computational capabilities. However, it is undoubted that when the model proposed by Menegotto and Pinto is used to contribute to the simulation of the structural behaviour of reinforced concrete members the inaccuracy in the prediction of the response of reinforcing steel bars is less important, i.e. the response of the cross-section is only partly influenced by steel. This last consideration may not be applied to steel structures, for which the model proposed by Menegotto and Pinto is also used. In these cases, the inaccuracy of the original model proposed by Menegotto and Pinto is expected to be significant if members are deformed in flexure and, particularly, if members are subjected to axial forces, as in buckling restrained braces.

Recently, Zsarnóczay and Budaházy [13] suggested some modifications to the model proposed by Menegotto and Pinto. Specifically, Zsarnóczay and Budaházy added a third asymptote with zero slope in order to model the eventual saturation of kinematic hardening and describe an ultimate perfectly plastic behaviour of steel [13]. Further, as hypothesized by Filippou et al. [12], they saved the parameters governing the response of all half-cycles. To simulate correctly the response of steel when it is partially unloaded and reloaded in the middle of a large amplitude cycle, they compared the stress at every load step to the stresses from previous cycles at the same strain level [13]. If the calculated stress exceeded the stress in the previous cycle by more than the isotropic hardening could account for, the parameters of the curve were modified so that the material behaviour would be described by the previous curve from that point on [13]. The Menegotto and Pinto model modified by Zsarnóczay and Budaházy [13] is implemented in OpenSees [8] as uniaxial material "Steel4".

3. The proposed model

The modification introduced by the writers is intended to eliminate the shortcoming observed by Filippou et al. [12], i.e. the proposed formulated is such that the steel response tends to the curve that would have developed if no unloading and reloading had occurred (this curve is later named *basic loading curve*). However, this modification does not require that the parameters of all performed half-cycles be saved and thus leads to a uniaxial material model which remains competitive with the original Menegotto and Pinto model in terms of the required calculus memory. To introduce the proposed modification, it is important to note that the uniaxial material response described by Menegotto and Pinto consists of two contributions. Fig. 4 shows these contributions in terms of normalized strain and stress. The first contribution to the normalized stress is given by a transition curve between two straight-line asymptotes (Fig. 4a); the first asymptote is characterized by a slope equal to $1 - b$, where b is the strain hardening ratio and the second is characterized by a null slope. The second contribution to the normalized stress is linear with the normalized strain and characterized by a slope equal to b (Fig. 4b). In particular, note that the total normalized stress σ^* is equal to unity at a strain ε^* equal to unity and that the second contribution (Fig. 4b) is much smaller than the first (Fig. 4a) because of the common values of the strain hardening of steel. Unlike the curve proposed by Menegotto and Pinto, the one suggested by the writers is a transition curve from a first straight line asymptote to the basic loading curve. The proposed curve in terms of the normalized strains and stresses may still be thought as the sum of two functions. The points selected for the normalization of strains and stresses are however different from those suggested by Menegotto and Pinto [1], as well as the analytical relations of the abovementioned stress

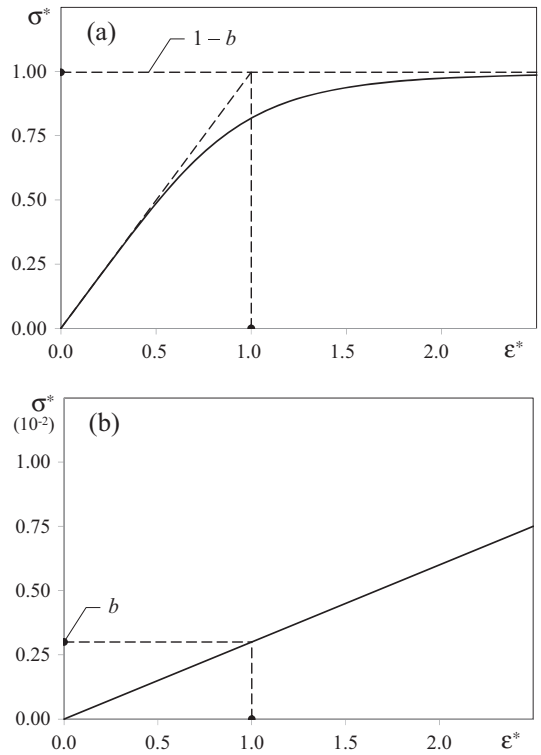

Fig. 4. Contributions to the transition curve proposed by Menegotto and Pinto in terms of normalized strain and stress.

contributions. In particular, the definition of point $P_0(\varepsilon_0,\sigma_0)$ has been changed with respect to Menegotto and Pinto. To construct a transition curve able to tend to the basic loading curve it is opinion of the writers that point P_0 should be defined as the intersection of the first asymptote to the basic loading curve. The new point P_0 is named here P_0' so as to differentiate it from the point P_0 considered by Menegotto and Pinto (see Fig. 5). The analytical evaluation of the coordinates of P_0' is not straightforward because of the nonlinearity of Eq. (5). Owing to this, they are derived by means of a trial-and-error method, verifying that the inclination of the line from the beginning of the reloading path to P_0' is equal to the Young's modulus of steel.

The proposed first contribution to the total normalized stress is given by a transition curve between two straight-line asymptotes; the first asymptote is characterized by a slope equal to $1 - e_p^*$ and

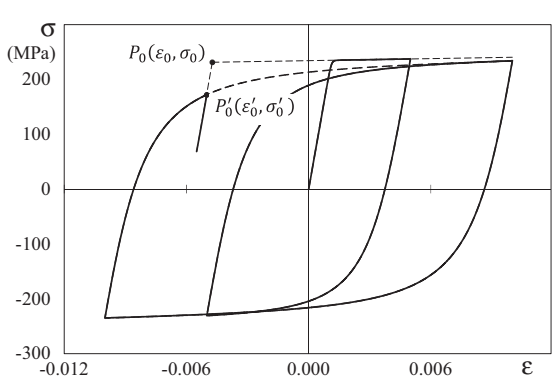

Fig. 5. Identification of point P_0'.

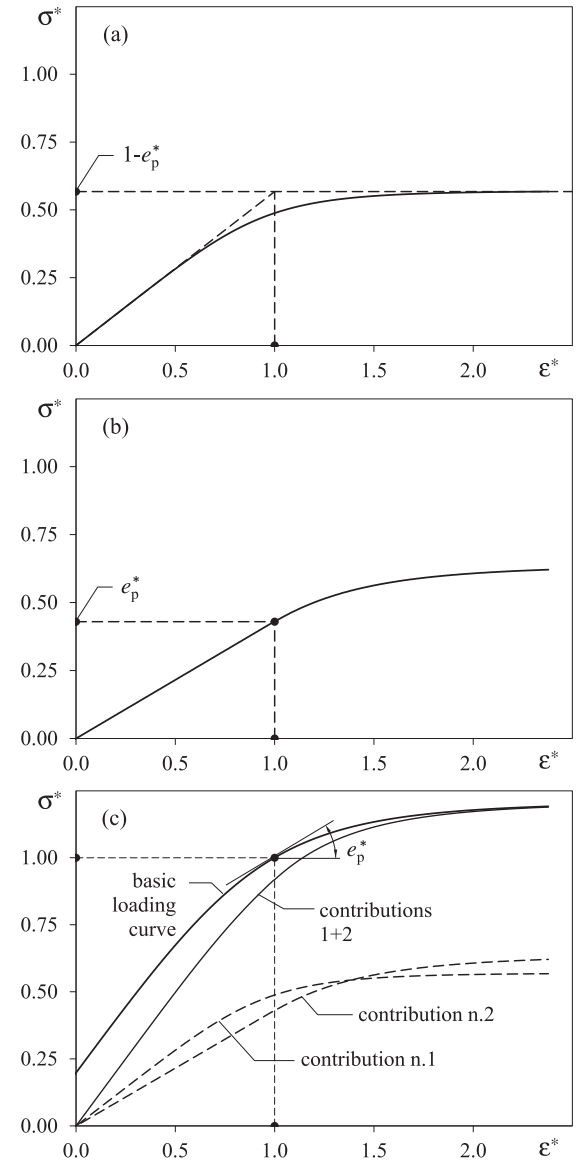

Fig. 6. Contributions to the proposed transition curve in terms of normalized strain and stress.

the second by a null slope (Fig. 6a). The parameter e_p^* is the slope of the basic loading curve in the reference system $(0, \varepsilon^*, \sigma^*)$ at a unity normalized strain, i.e. in P_0'. The transition curve proposed by the writers is

$$\sigma^* = \frac{(1 - e_p^*)\varepsilon^*}{(1 + \varepsilon^{*R})^{\frac{1}{R}}} \tag{8}$$

This relation is similar to that suggested by Menegotto and Pinto [see Eq. (5)] save that the strain hardening ratio b is substituted for e_p^*. The parameter R in Eq. (8) is that of the basic loading curve.

Fig. 6b shows the second contribution to the normalized stress. The branch of the curve with normalized strains not higher than unity is linear and characterized by a slope equal to e_p^*. The remaining part of the curve ($\varepsilon^* > 1$), instead, is defined by a nonlinear function and takes account of the nonlinearity of the basic loading curve. The analytical relations of the two parts of this second contribution are

$$\sigma^* = e_p^* \, \varepsilon^* \qquad \varepsilon^* \leqslant 1 \tag{9}$$

$$\sigma^* = f_b(\varepsilon^*) - (1 - e_p^*) \qquad \varepsilon^* > 1 \tag{10}$$

where $f_b(\varepsilon^*)$ is the normalized stress of the basic loading curve in terms of the normalized strain.

Thus, the proposal for the transition curve in the case of reloading after partial unloading is

$$\sigma^* = e_p^* \varepsilon^* + \frac{\left(1 - e_p^*\right)\varepsilon^*}{(1 + \varepsilon^{*R})^{\frac{1}{R}}} \qquad \varepsilon^* \leqslant 1 \tag{11}$$

$$\sigma^* = f_b(\varepsilon^*) - \left(1 - e_p^*\right) + \frac{\left(1 - e_p^*\right)\varepsilon^*}{(1 + \varepsilon^{*R})^{\frac{1}{R}}} \qquad \varepsilon^* > 1 \tag{12}$$

The true (non normalized) stress is

$$\sigma = \sigma^*(\sigma_0 - \sigma_r) + \sigma_r \tag{13}$$

4. Implementation

The model has been implemented in the program OpenSees [8] by changing the uniaxial material model "Steel02" [18]. Like the latter model, the one proposed in this paper is defined by assigned values of the yield stress σ_0, yield strain ε_0, Young's modulus E, kinematic hardening ratio b, coefficients R_0, a_1, a_2, a_3 and a_4. At the beginning of the analysis the plastic excursion is null and then the parameter R that characterises the roundness of the loading curve is equal to R_0. The formulation proposed by Menegotto and Pinto [1] and modified by Filippou et al. [12] is utilized in case loading starts from the maximum or minimum strain. In all the other cases, the relation proposed in Eq. (8) is used. To apply the proposed loading curve, the values of the parameters that characterise the basic loading curve are recorded during the analysis. This holds for the basic curve corresponding to either increments or decrements of strain.

Note that the parameter e_p^* in Eqs. (8–12) is the slope of the basic loading curve in P_0' and that it is calculated with respect to stresses and strains that are normalized to the differences between stresses and strains of points P_0' and P_r of the reloading curve. So, once point P_0' has been identified, three subsequent operations are required to obtain the value of the parameter e_p^*. The slope of the basic loading curve is first determined as normalized to the stresses and strains of points P_0^b and P_r^b of the basic loading curve (Fig. 7)

$$e_p^{*,b} = b - \frac{(1 - b)}{\left[1 + (\varepsilon_0'^*)^R\right]^{\frac{1}{R}}} \tag{14}$$

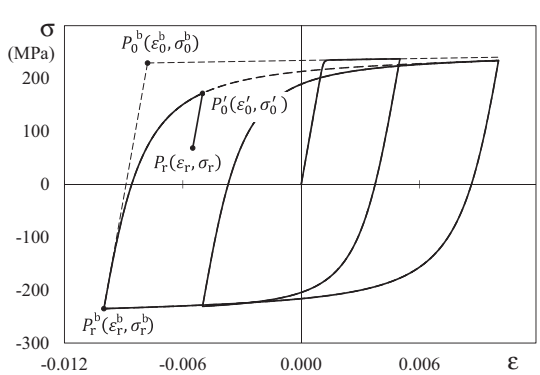

Fig. 7. Definition of points P_0^b and P_r^b.

Then, the slope is denormalized, i.e. expressed in terms of non-normalized stresses and strains

$$e_p = e_p^{*,b} \frac{\left(\sigma_0^b - \sigma_r^b\right)}{\left(\varepsilon_0^b - \varepsilon_r^b\right)} \tag{15}$$

Finally, the slope is calculated as a function of stresses and strains that are normalized to those of points P_0' and P_r of the reloading curve

$$e_p^* = e_p \frac{(\varepsilon_0' - \varepsilon_r)}{(\sigma_0' - \sigma_r)} \tag{16}$$

Similarly, the function $f_b(\varepsilon^*)$ of Eq. (10) must be normalized to stresses and strains of points P_0' and P_r of the reloading curve. To this end, the basic loading curve (normalized to stresses and strains of points P_0^b and P_r^b) is first denormalized and then normalized to stresses and strains of points P_0' and P_r.

5. Simple numerical applications

To show the effects of the suggested corrections, the new model proposed by the authors and the uniaxial material model Steel4 are applied to a steel bar subjected to cyclic axial strain according to four assigned load histories (Fig. 8). The response of the new model and that of the material model Steel4 are shown by thick line in Figs. 9 and 10, respectively. In the same figures, the basic loading curve is plotted by dashed line. The values assigned to the parameters of the models are σ_0 = 235 MPa, E_0 = 210,000 MPa (and thus ε_0 = 0.112%), R_0 = 20, a_1 = 18.5, a_2 = 0.15, a_3 = 0 and a_4 = 1. The values of the last two parameters (a_3 and a_4) are not required when the Steel4 material is adopted.

In Fig. 9a, a single partial unloading is considered within the load history (Fig. 8a). The comparison between the thick and the dashed lines shows the effectiveness of the proposed formulation. The improvement with respect to the original model stands out from the comparison of Fig. 9a with Fig. 3b where the same load history is considered. In the other cases (Fig. 9b–d) two partial unloading paths are applied (Fig. 8b–d). Specifically, in the second case (Fig. 9b), the unloading paths (UP) begin at points labeled UP_1 and UP_2 in the figure. Both the reloading curves are built with reference to equal values of the parameters of the basic loading curve. In fact, these values are updated only if partial unloading takes place at strains greater than the maximum recorded value (or lower than the minimum value). Owing to this, both the reloading curves tend to the same basic loading curve. A more refined formulation of the model would require an updating of the parameters of the basic loading curve at each loading reversal. However, the analytical relations (Eqs. (9) and (10)) of the two parts of the second contribution to the proposed stress-strain relationship would become more and more complex. Further, a greater computation effort would be required to record the values of the parameters that characterise the basic loading curve at each loading reversal.

In the response represented in Fig. 9c the reloading curves tend to the same basic loading curve but the flaw is negligible because an updating of the values of the parameters of the reloading curve would only slightly change the slope of this curve. Instead, the flaw highlighted in Fig. 9b does not occur in the last case (Fig. 9d). In fact, in this case the reloading curves tend to two different basic loading curves.

Finally, based on the results shown in Fig. 10, the uniaxial material model Steel4 appears to be able to simulate correctly the cyclic response of steel only for two of the four considered loading histories (Fig. 10a and d).

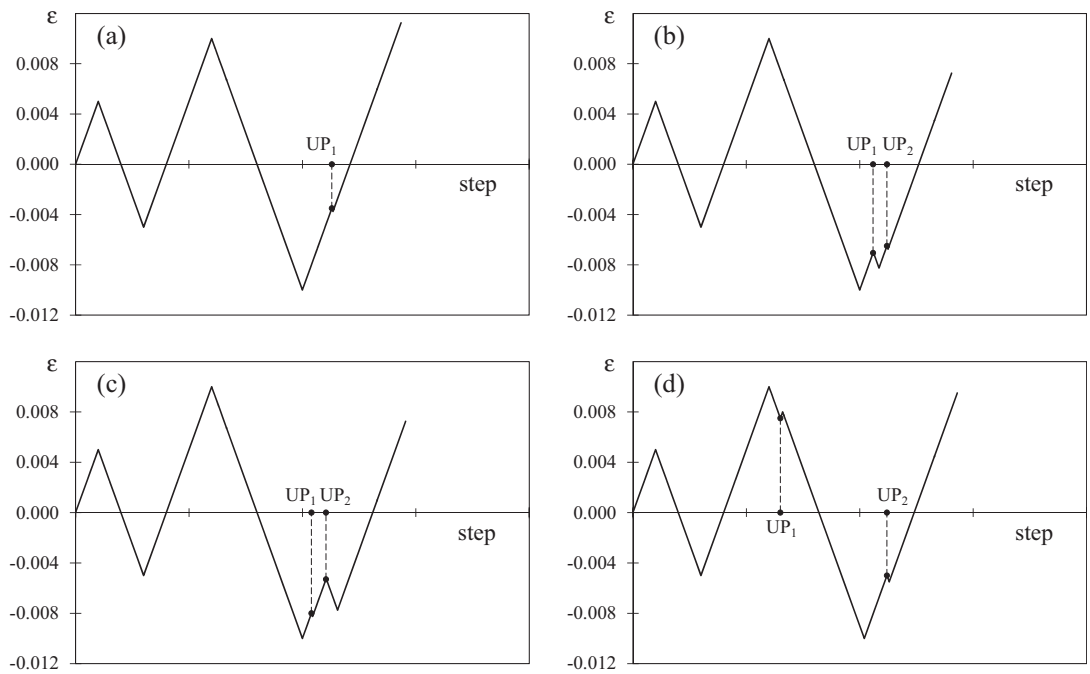

Fig. 8. Axial loading histories.

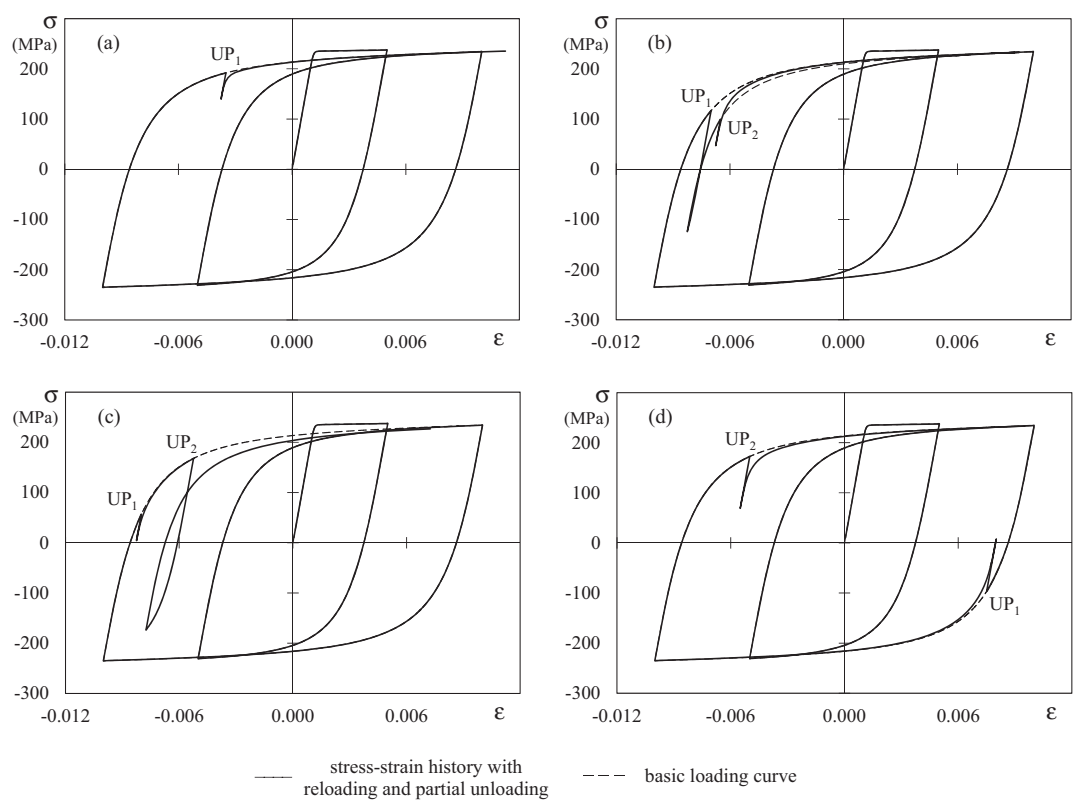

Fig. 9. The improved model in the case of reloading after partial unloading.

6. Numerical analyses

In this section, five multi-storey buildings are considered. The first two structures are a six-storey r.c. frame and a six-storey steel moment resisting frame. The other three structures are eight-storey braced frames and are equipped with buckling restrained braces, concentrically chevron (conventional) braces or tension only braces.

The seismic response of these frames is analysed by nonlinear dynamic analysis by means of the OpenSees program. Attention is focused on the moment–curvature time-history at the ends of the members and on the axial force-axial deformation time-

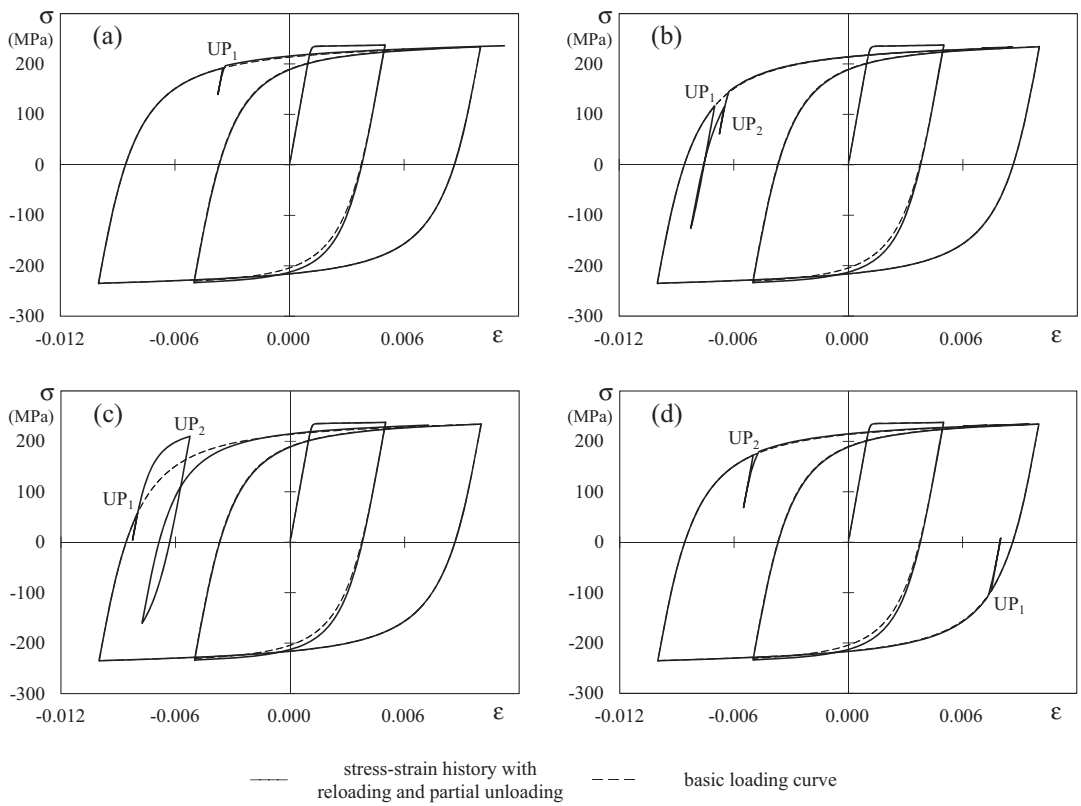

Fig. 10. The model proposed by Zsarnóczay and Budaházy [13] in the case of reloading after partial unloading.

history of braces. Global response parameters are also considered in terms of the maximum storey drifts.

6.1. Design of the frames

Fig. 11 illustrates the geometric scheme of the r.c. frame. The frame is designed to sustain gravity loads and seismic actions according to the provisions reported in Eurocode 8 for high ductility r.c. structures (behaviour factor equal to 5.85). The characteristic value of the compressive cylinder strength of concrete f_{ck} is 25 MPa and the yield stress of reinforcing steel f_{yk} is 450 MPa. Two load conditions are considered for the evaluation of the design internal forces. The first load condition considers the characteristic values of dead and live loads (G_k and Q_k), increased by the partial safety factors γ_g and γ_q; the second considers seismic actions, characteristic dead loads (G_k) and quasi-permanent live loads (ψ_2 Q_k). The gravity loads applied to the beams and columns of the r.c. structure in the first load condition are q = 54.9 kN/m, F_1 = 88.2 kN and F_2 = 42.7 kN; those considered in the second load condition are q = 31.5 kN/m, F_1 = 63.0 kN and F_2 = 30.5 kN. The mass of each storey is equal to 57.6 t. The base shear is calculated assuming the elastic response spectrum proposed for soft soil (soil C according to Eurocode 8) and characterized by a peak ground acceleration equal to 0.35 g. The design seismic forces are distributed in elevation according to an inverted triangular distribution.

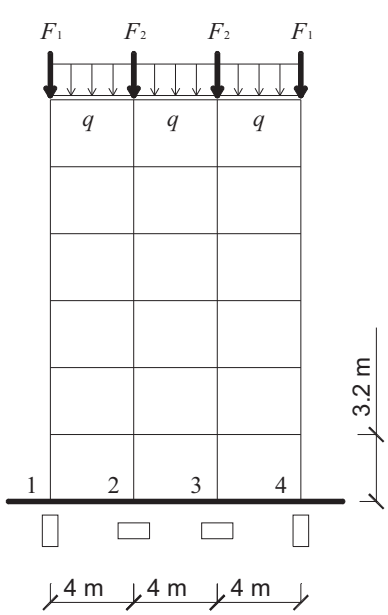

Fig. 11. Analysed r.c. frame.

The rectangular cross-sections of the beams are characterized by a width b equal to 30 cm; the depth h is equal to 50 cm at the two upper storeys and 60 cm at the other storeys. The cover to the longitudinal reinforcement c is equal to 4 cm. Table 1 lists the top and bottom longitudinal reinforcements ($A_{s,t}$ and $A_{s,b}$). The column cross sections are 30×60 cm^2 and are oriented as shown in Fig. 11. The total longitudinal reinforcement of each column is equal to 24.92 cm^2 and that on each short side of the cross-section is 9.42 cm^2.

The steel moment resisting frame (Fig. 12) is designed according to the procedure proposed in Neri [19] and is described in detail in Ghersi et al. [20] as case 5DDL. The storey mass is equal to 43.6 t. The gravity loads applied to the beams are equal to 13.0 kN/m. European wide-flange sections and steel grade S275 (yield stress f_y equal to 275 MPa) are used for beams and columns. The adopted cross sections are summarised in Table 2.

Table 1
Longitudinal reinforcement of the beams of the r.c. frame.

Storey	Section	Beam 1–2				Beam 2–3				Beam 3–4			
		$A_{s,t}^{(left)}$ (cm^2)	$A_{s,b}^{(left)}$ (cm^2)	$A_{s,t}^{(right)}$ (cm^2)	$A_{s,b}^{(right)}$ (cm^2)	$A_{s,t}^{(left)}$ (cm^2)	$A_{s,b}^{(left)}$ (cm^2)	$A_{s,t}^{(right)}$ (cm^2)	$A_{s,b}^{(right)}$ (cm^2)	$A_{s,t}^{(left)}$ (cm^2)	$A_{s,b}^{(left)}$ (cm^2)	$A_{s,t}^{(right)}$ (cm^2)	$A_{s,b}^{(right)}$ (cm^2)
6	30 × 50	4.71	4.71	6.28	4.71	6.28	4.71	6.28	4.71	6.28	4.71	4.71	4.71
5	30 × 50	7.82	4.71	9.42	6.28	9.42	6.28	9.42	6.28	9.42	6.28	7.82	4.71
4	30 × 60	7.82	6.28	10.97	7.82	10.97	7.82	10.97	7.82	10.97	7.82	7.82	6.28
3	30 × 60	9.42	6.28	14.11	9.42	14.11	9.42	14.11	9.42	14.11	9.42	9.42	6.28
2	30 × 60	10.97	7.82	15.71	10.97	15.71	10.97	15.71	10.97	15.71	10.97	10.97	7.82
1	30 × 60	10.97	7.82	15.71	10.97	15.71	10.97	15.71	10.97	15.71	10.97	10.97	7.82

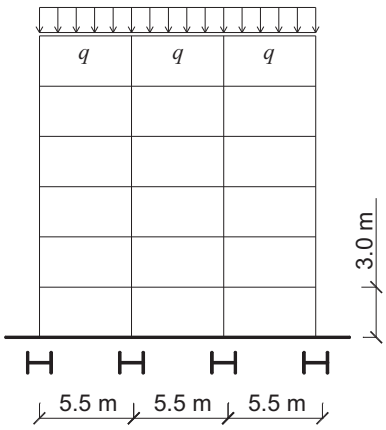

Fig. 12. Analysed steel moment resisting frame.

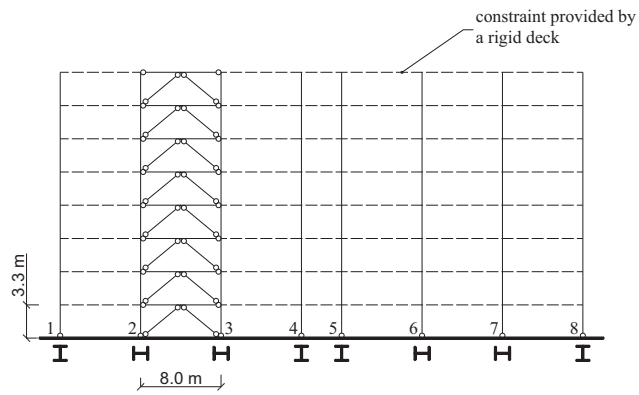

Fig. 13. Analysed buckling resistant and conventional concentrically braced frames.

Table 2
Cross-section of the steel moment resisting frame.

Storey	Beams	Columns
6	IPE 300	HEB 240
5	IPE 300	HEB 260
4	IPE 300	HEB 260
3	IPE 300	HEB 260
2	IPE 300	HEB 260
1	IPE 300	HEB 300

Steel grade S 275.

Table 3
Cross-section area of the yielding core of the braces and section of the beams and columns (q = 4).

Storey	Buckling restrained braced frame			Gravity columns		
	A_c (cm^2)	Beams	Columns	Columns 1, 4	Columns 5, 8	Columns 6, 7
8	9.36	HEA 180	HEB 160	HEB 160	HEB 160	HEB 200
7	14.63	HEA 180	HEB 160	HEB 160	HEB 160	HEB 200
6	18.03	HEA 200	HEB 240	HEB 200	HEB 240	HEB 280
5	20.71	HEA 200	HEB 240	HEB 200	HEB 240	HEB 280
4	23.34	HEA 200	HEB 300[a]	HEB 240	HEB 300[a]	HEB 360
3	26.03	HEA 220	HEB 300[a]	HEB 240	HEB 300[a]	HEB 360
2	28.46	HEA 220	HEB 450[a]	HEB 280	HEB 450[a]	HEB 400[a]
1	30.03	HEA 220	HEB 450[a]	HEB 280	HEB 450[a]	HEB 400[a]

Steel grade S 235.
[a] Steel grade S 275.

The buckling restrained braced frame (Fig. 13) is designed according to the procedure proposed by Bosco and Marino [21], assuming a design storey drift equal to 1.25% times the interstorey height. The frame model also comprises six columns that sustain only gravity loads. The span length is 8 m and the interstorey height is 3.3 m. The length L_c of the yielding core of the buckling restrained brace (BRB) is 0.5 times the work-point to work-point length L_w of the brace. The length of the connection segment is $L_j/2 = 0.65$ m whereas that of the transition segment is $L_t/2 = 0.5$ $(L_w - L_c - L_j)$. The cross-section areas of the connection and transition segments (A_j and A_t) are such that $A_c/A_t = 0.5$ and $A_c/A_j = 0.3$, where A_c is the cross-section of the yielding core. The storey mass is 146.8 t. Steel grade S 235 ($f_y = 235$ MPa) is used for all the braces. The cross-section area A_c is that strictly required by design. Table 3 reports the cross-section area of the yielding core of the braces and the section of the beams and columns. More details may be found in Bosco and Marino [21].

The two frames with conventional braces (Figs. 13 and 14) are designed according to Eurocode 8. The geometry of these frames (number of storeys, interstorey height and span length) is equal to that of the frame with BRBs. The storey mass is 146.8 t in the building with chevron braces and 244.65 t in that with tension only braces. Hollow squared sections are used for braces while wide flange sections are used for beams and columns (Table 4). More details on these frames may be found in Marino [22].

6.2. Modelling of the frames

The structural members of the r.c. frame and those of the steel moment resisting frame are modelled by means of *Beam With Hinge Elements* [23]. This type of elements has been preferred to elements with distributed plasticity to limit the computational effort. This choice is reasonable because, in the occurrence of the ground motion, only a small portion of the gravity loads considered in the non-seismic design situation is supposed to be applied to the beams and thus the formation of a beam span plastic hinge is not expected. The uniaxial model referred to as Concrete 04 in Opensees has been used for concrete. For loading in compression, the envelope to the stress-strain curve follows the model proposed by Popovics [24]. The relationships proposed by Mander et al.

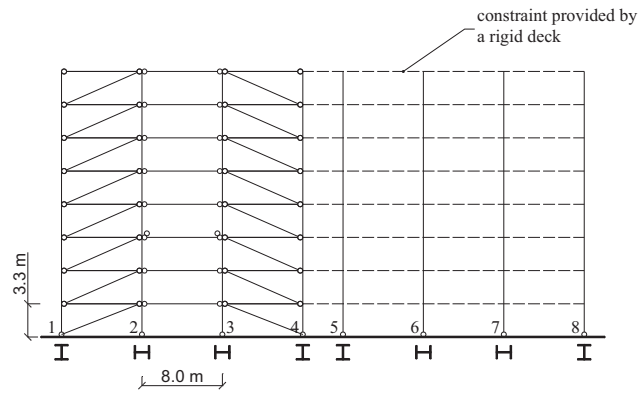

Fig. 14. Analysed frames with tension only braces.

Table 4
Section of the braces, beams and columns of the conventional braced frame.

	Chevron braced frame			Tension only braced frame		
Storey	Braces	Beams	Columns	Braces	Beams	Columns
8	S 135 × 5	HEA 550	HEB 180	S 135 × 4	HEA 240	HEB 180
7	S 150 × 6	HEB 600	HEB 180	S 160 × 5	HEA 240	HEB 180
6	S 175 × 6	HEA 650	HEB 240	S 160 × 6	HEA 240	HEB 240
5	S 180 × 6	HEA 700	HEB 240	S 175 × 6	HEA 240	HEB 240
4	S 200 × 6	HEA 700	HEB 300[a]	S 180 × 6	HEA 240	HEB 280[a]
3	S 200 × 7	HEA 800	HEB 300[a]	S 200 × 7	HEA 240	HEB 280[a]
2	S 220 × 7	HEA 800	HEB 450[a]	S 220 × 7	HEA 240	HEB 360[a]
1	S 220 × 7	HEA 800	HEB 450[a]	S 220 × 7	HEA 240	HEB 360[a]

Steel grade S 235.
[a] Steel grade S 275.

[25] are used to calculate the confinement effect on the concrete monotonic behaviour. For unloading and reloading in compression, the Karsan-Jirsa model [26] is used to determine the slope of the curve. For tensile loading, an exponential curve is used to define the envelope to the stress-strain curve. For unloading and reloading in tension, the secant stiffness is used to define the path.

The stress–strain relationship of the reinforcing steel is separately modelled by means of the original formulation of the model by Menegotto and Pinto and by the proposed version of the model above.

The beams and columns of the braced frames are considered as elastic members. Buckling restrained braces are modelled by means of three elements. The ending elements (denoted 1 and 3 in the equations below) are elastic and schematize the connection and transition segments of the brace. The length and cross-sectional area of each of these elements are given by the following relations

$$L^1 = L^3 = \frac{L_j + L_t}{2} \tag{17}$$

$$A^1 = A^3 = \frac{A_c(L_j + L_t)}{L_j \frac{A_c}{A_j} + L_t \frac{A_c}{A_t}} \tag{18}$$

The central element simulates the yielding core and is modelled by a "*nonlinear beam Column*" element [27–31]. The strain-hardening ratio (ratio of the post-yield stiffness to the initial elastic stiffness) is assigned so as to include the effects of both kinematic and isotropic hardening.

The conventional braces are modelled by four "*nonlinear Beam Column*" elements. The cross-section is divided into 20 fibres. An initial camber displacement of 0.1% of the brace length is applied at brace mid-length. The corotational theory is used to represent the moderate to large deformation effects of inelastic buckling of the brace [32]. The original formulation of the model by Menegotto and Pinto and the proposed version of the same model are separately considered for the response of conventional and buckling restrained braces.

6.3. Seismic input

Two sets of accelerograms are considered as the seismic input. The first set consists of the 20 ground motions adopted in the FEMA/SAC project [33]. These ground motions represent seismic events with a 10% probability of exceedance in 50 years in the US Los Angeles area. The second set consists of ten artificially generated accelerograms characterized by a total duration of 30.5 s and a length of the stationary part equal to 7 s [34]. The response spectra of the accelerograms of the latter set are compatible with the elastic response spectrum reported in Eurocode 8 for soft soil (Soil Type C). The peak ground acceleration is equal to 0.35 g. The response spectra of the single accelerograms of the two sets are shown in Fig. 15 by thin grey lines. A black line shows the average response spectrum. In the same figure the fundamental periods of vibration of the designed structures are marked.

6.4. Detailed seismic response of the buckling restrained braced frame

The seismic response of the buckling restrained braced frame subjected to one natural accelerogram and one artificial accelerogram is thoroughly discussed in this section. Out of the designed structures, the frame with BRBs has been selected because its response is mainly dictated by the axial behaviour of braces. Thus, all fibres of a brace are subjected to the same stress-strain cyclic behaviour.

Fig. 16 shows a comparison between the response predicted by the Menegotto Pinto model implemented in OpenSees (represented by a dashed line) and that given by the proposed model (solid line). Specifically, Fig. 16a reports the axial force-axial displacement time-history of the BRB of the 8th-storey of the structure subjected to the natural accelerogram whereas Fig. 16b shows that of the BRB belonging to the 5th-storey of the structure subjected to the artificial accelerogram. The figure confirms that the proposed modifications are able to correct the flaw of the cyclic behaviour of the original version of the Menegotto Pinto model. The maximum brace elongation and shortening produced by the proposed model ($\Delta L^{P,+}_{max}$ and $\Delta L^{P,-}_{max}$) and by the original version of the Menegotto Pinto ($\Delta L^{O,+}_{max}$ and $\Delta L^{O,-}_{max}$) are also marked. The figure shows that the maximum axial displacements predicted by the two models can be different. These differences depend on the loading histories. Thus, compared with the modified Menegotto Pinto model, the original model can lead to conservative results ($\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max} < 1$, $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max} < 1$ as in Fig. 16a), unconservative results ($\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max} > 1$ in Fig. 16b) or similar results. To assess the differences in the responses provided by the two versions of the Menegotto Pinto model, the following procedure has been followed. (1) For each accelerogram and for each brace, the ratio $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$ between the maximum elongation predicted by the proposed model and that predicted by the original model is calculated. Similarly, the ratio is calculated for the maximum shortening ($\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$) and for the absolute values of the axial displacements $\Delta L^{P}_{max}/\Delta L^{O}_{max}$, where $\Delta L_{max} = \max\{\Delta L^{+}_{max}, -\Delta L^{-}_{max}\}$. (2) Of the ratios obtained for all the braces and accelerograms, the maximum values (max $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$, max $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$, max $\Delta L^{P}_{max}/\Delta L^{O}_{max}$) and the minimum values (min $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$, min $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$, min $\Delta L^{P}_{max}/\Delta L^{O}_{max}$) are recorded and reported in Table 5. These values highlight that the differences between the

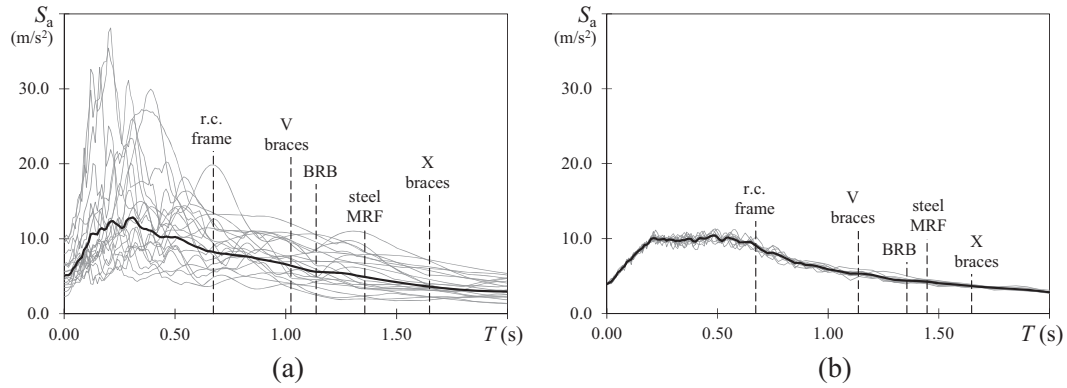

Fig. 15. Periods of the designed structures and response spectra for (a) natural accelerograms (b) artificial accelerograms.

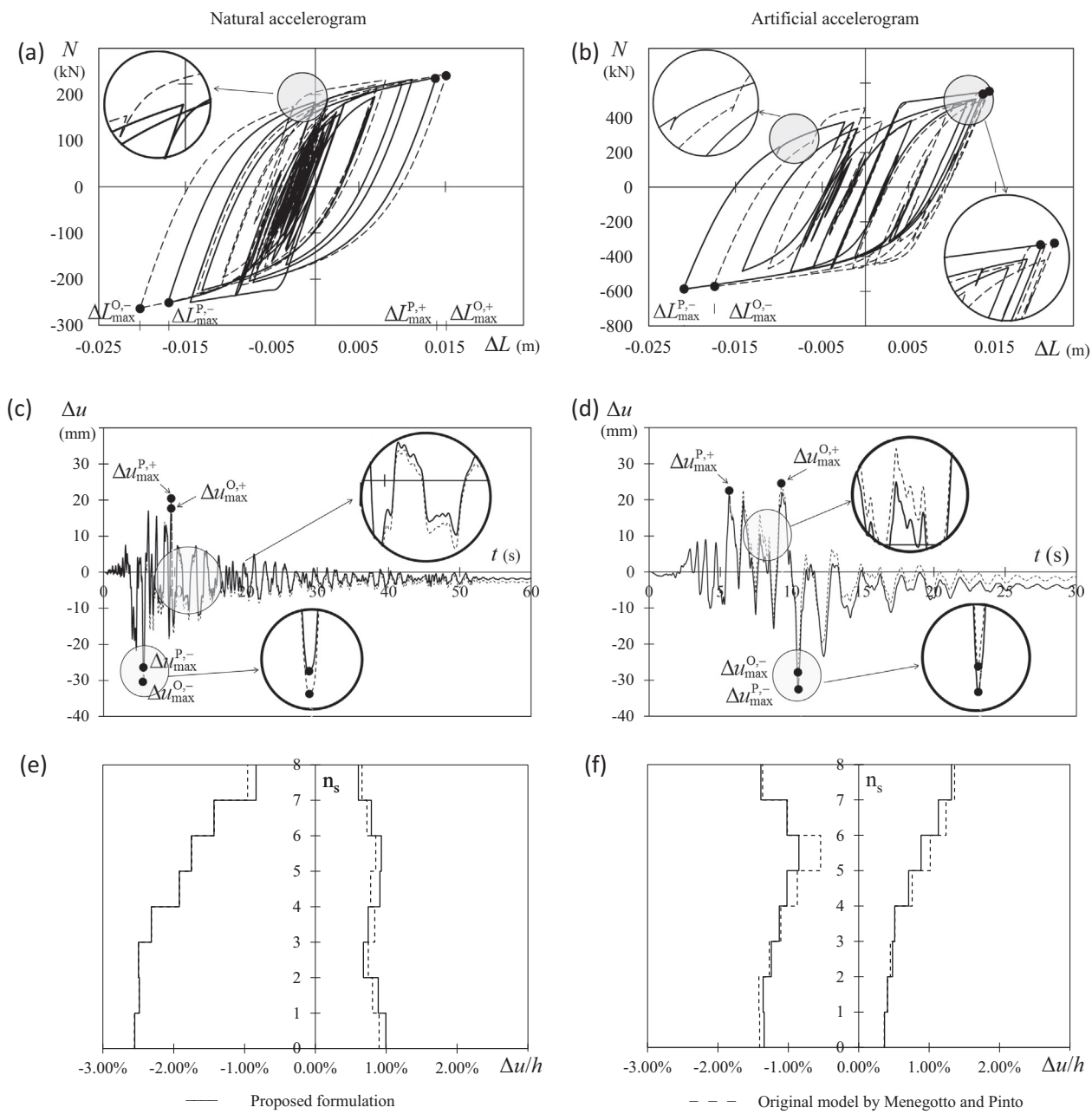

Fig. 16. Response of the frame with BRB: (a and b) axial force-axial displacement time-histories; (c and d) storey drifts time-histories; (e and f) heightwise distribution of the maximum storey drift angles.

results of the two steel models are generally fairly large. In particular, the values of the ratios $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$ and $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$ are in the range 0.65–2.1 when referring to the natural accelerograms and in the range 0.75–1.7 when referring to the artificial accelerograms. The ratios $\Delta L^{P}_{max}/\Delta L^{O}_{max}$ are in the range 0.705–1.497.

The differences between the results provided by the two models of steel have also been calculated for the storey drift Δu. Fig. 16c reports the storey drift time-history at the 8th storey of the structure subjected to the natural accelerogram whereas Fig. 16d reports the storey drift time-history at the 5th storey of the structure subjected to the artificial accelerogram. These figures show that the differences in the time histories increase after some partial unloading and reloading and that they may affect not only the maximum ($\Delta u^{P,+}_{max}$ and $\Delta u^{O,+}_{max}$) or minimum ($\Delta u^{P,-}_{max}$ and $\Delta u^{O,-}_{max}$) drift but also the residual drift. Further, the maximum predicted drifts can be reached at different times (as an example, see $\Delta u^{P,+}_{max}$ and $\Delta u^{O,+}_{max}$ in Fig 16d). The drifts are processed according to a procedure similar to that adopted for the axial deformations of braces. The obtained results are reported in Table 5. As expected, the maximum differences in the storey drifts produced by the two models are slightly lower than those recorded for the elongation/shortening of braces. However, they still remain fairly large.

The heightwise distribution of the maximum storey drift angles (i.e. the maximum storey drifts normalized to the interstorey height h) produced by the two considered accelerograms is plotted in Fig. 16e and f. These figures show that, despite the non-negligible differences in the maximum storey drifts, both models lead to a similar prediction of the global response of the structure.

Finally, the two versions of the steel model are compared in terms of the required computational effort, i.e. in terms of the time required by the structural analysis. In this regard, the authors note that the proposed modifications cause a small increase in the computational time. For the structure under examination, the increase is about 5%. The increase is probably due to the small tolerance accepted for the iterative calculation of point P'_0. Further, at this stage the procedure is not optimized to reduce the time of analysis.

To validate further the proposed model, the response of the frame with BRBs has also been determined by means of the uniaxial material model Steel4. In this regard, the parameter that controls the number of half-cycles saved during the loading history is set equal to the default value in OpenSees (i.e. 50 cycles). The obtained results are reported in Fig. 17 where the response predicted by the original Menegotto Pinto model is (once again) represented by a dashed line, while that given by the uniaxial material model Steel4 is represented by a solid line. Fig. 17a and b confirms that the material model Steel4 is only sporadically able to simulate the cyclic response correctly when partial reloading after unloading occurs.

6.5. Seismic response of all the considered structures

In this section, the responses predicted by the original version of the Menegotto Pinto material and by the proposed model for all the considered structures are synthetically compared. The differences in the responses provided by the two versions of the steel model in the members of the moment resisting frames are compared in terms of the curvature at the ends of beams and columns. For each accelerogram, the maximum positive value and the minimum negative value of the curvatures (χ^{+}_{max} and χ^{-}_{max}) are calculated at the ends of beams and columns of every storey. For each accelerogram and for each end of the members, the ratio between the maximum positive curvature predicted by the proposed model and that predicted by the original model is calculated ($\chi^{P,+}_{max}/\chi^{O,+}_{max}$). Similarly, the ratio is calculated for the negative values of the curvatures ($\chi^{P,-}_{max}/\chi^{O,-}_{max}$) and for the absolute values of the maximum curvatures, either positive or negative, ($\chi^{P}_{max}/\chi^{O}_{max}$). Of the ratios obtained for all the members and accelerograms, the maximum values (max $\chi^{P,+}_{max}/\chi^{O,+}_{max}$, max $\chi^{P,-}_{max}/\chi^{O,-}_{max}$, max $\chi^{P}_{max}/\chi^{O}_{max}$) and the minimum values (min $\chi^{P,+}_{max}/\chi^{O,+}_{max}$, min $\chi^{P,-}_{max}/\chi^{O,-}_{max}$, min $\chi^{P}_{max}/\chi^{O}_{max}$) are recorded and reported in tables. Specifically, Table 6 refers to the results relative to the natural accelerograms considered while Table 7 refers to the artificial accelerograms.

The analysis of the seismic response of the r.c. frame proves that the behaviour of beams and columns of this frame is virtually independent of the model adopted for steel. In general (Tables 6 and 7), the maximum ratios $\chi^{P,+}_{max}/\chi^{O,+}_{max}$ and $\chi^{P,-}_{max}/\chi^{O,-}_{max}$ of the beams are slightly lower than those of the columns (e.g., for natural accelerograms, max $\chi^{P,+}_{max}/\chi^{O,+}_{max}$ = 1.108 and max $\chi^{P,-}_{max}/\chi^{O,-}_{max}$ = 1.098 in beams while max $\chi^{P,+}_{max}/\chi^{O,+}_{max}$ = 1.187 and max $\chi^{P,-}_{max}/\chi^{O,-}_{max}$ = 1.224 in columns). The ratios $\chi^{P}_{max}/\chi^{O}_{max}$ are in the range from 0.846 to 1.128.

The differences between the responses provided by the two models become more significant in the steel moment resisting frame. In this case, the maximum ratios $\chi^{P,+}_{max}/\chi^{O,+}_{max}$ and $\chi^{P,-}_{max}/\chi^{O,-}_{max}$ of the beams are as high as 1.80 while those of columns are higher than 1.30. Similarly, the minimum ratios $\chi^{P,+}_{max}/\chi^{O,+}_{max}$ and $\chi^{P,-}_{max}/\chi^{O,-}_{max}$ of beams and columns may be lower than 0.83 and 0.81. The ratios between the absolute maximum demands $\chi^{P}_{max}/\chi^{O}_{max}$ are in the range from 0.807 to 1.256 (see Tables 6 and 7).

The procedure described in the previous section is used to compare the maximum elongation and shortening of braces of the two frames with conventional braces. The lower part of Tables 6 and 7 lists the maximum and minimum values of the ratios $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$, $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$ and $\Delta L^{P}_{max}/\Delta L^{O}_{max}$. All these values highlight that the differences between the results of the two steel models are generally fairly large in the examined braced frames. In particular, the differences are higher for the frame with conventional braces in

Table 5
Minimum and maximum values of the ratios $\Delta L^{P}_{max}/\Delta L^{O}_{max}$ and $\Delta u^{P}_{max}/\Delta u^{O}_{max}$ for the frame with BRBs.

	Positive values		Negative values		Absolute values	
	Min $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$	Max $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$	Min $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$	Max $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$	Min $\Delta L^{P}_{max}/\Delta L^{O}_{max}$	Max $\Delta L^{P}_{max}/\Delta L^{O}_{max}$
Natural	0.650	2.053	0.657	2.022	0.705	1.497
Artificial	0.754	1.728	0.754	1.716	0.806	1.201
	Min $\Delta u^{P,+}_{max}/\Delta u^{O,+}_{max}$	Max $\Delta u^{P,+}_{max}/\Delta u^{O,+}_{max}$	Min $\Delta u^{P,-}_{max}/\Delta u^{O,-}_{max}$	Max $\Delta u^{P,-}_{max}/\Delta u^{O,-}_{max}$	Min $\Delta u^{P}_{max}/\Delta u^{O}_{max}$	Max $\Delta u^{P}_{max}/\Delta u^{O}_{max}$
Natural	0.712	1.335	0.875	1.700	0.782	1.335
Artificial	0.766	1.221	0.853	1.572	0.831	1.168

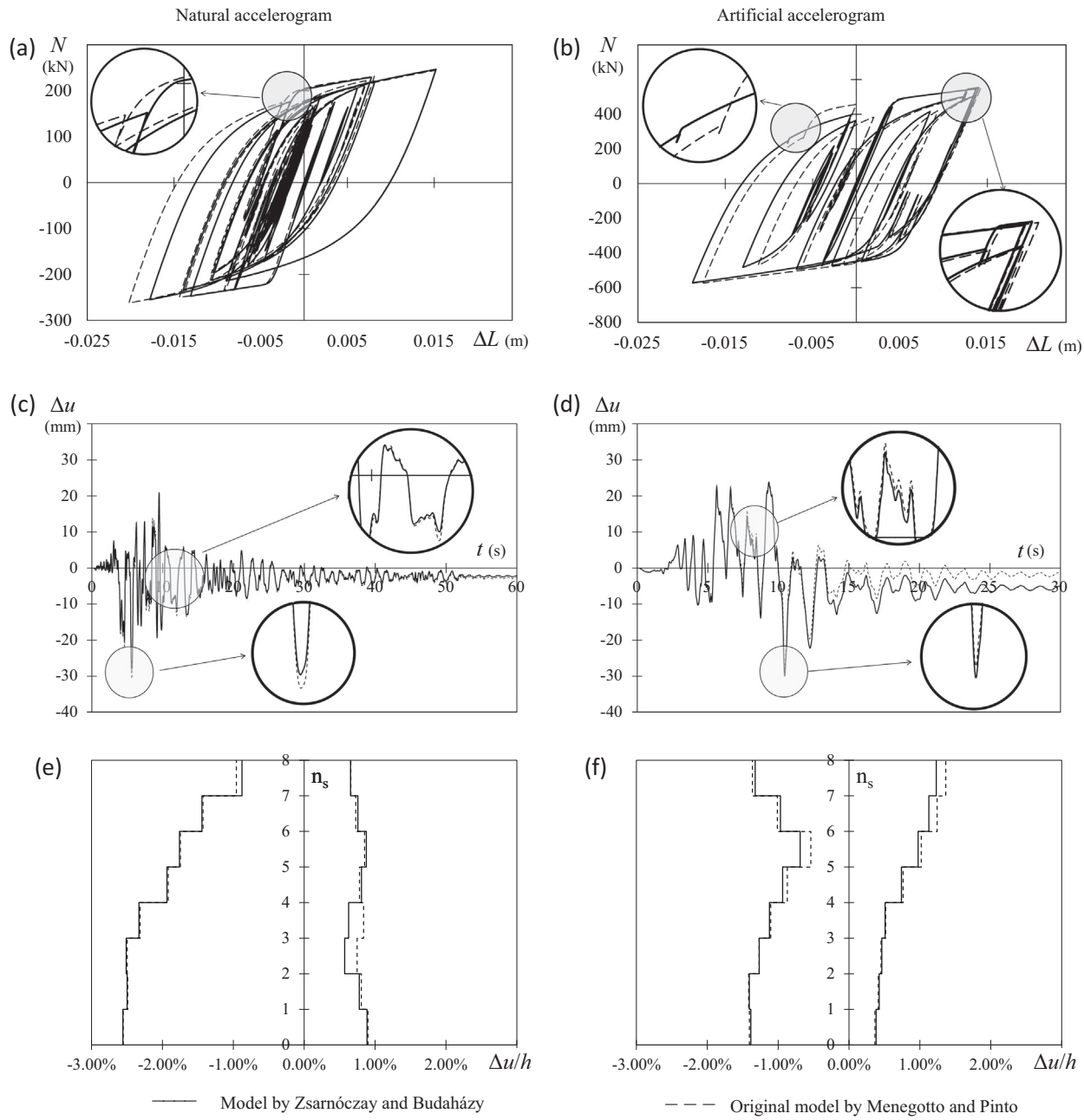

Fig. 17. Response of the frame with BRB: (a and b) axial force-axial displacement time-histories; (c and d) storey drifts time-histories; (e and f) heightwise distribution of the maximum storey drift angles.

Table 6
Minimum and maximum values of the ratios $\chi^{P}_{max}/\chi^{O}_{max}$, $\Delta L^{P}_{max}/\Delta L^{O}_{max}$ (natural accelerograms).

Frame	Members	Positive values		Negative values		Absolute values	
		Min $\chi^{P,+}_{max}/\chi^{O,+}_{max}$	Max $\chi^{P,+}_{max}/\chi^{O,+}_{max}$	Min $\chi^{P,-}_{max}/\chi^{O,-}_{max}$	Max $\chi^{P,-}_{max}/\chi^{O,-}_{max}$	Min $\chi^{P}_{max}/\chi^{O}_{max}$	Max $\chi^{P}_{max}/\chi^{O}_{max}$
r.c. frame	Beams	0.882	1.108	0.899	1.098	0.882	1.098
	Columns	0.854	1.187	0.846	1.224	0.846	1.125
Steel MRF	Beams	0.828	1.800	0.872	1.430	0.872	1.256
	Columns	0.849	1.225	0.807	1.323	0.807	1.115
		Min $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$	Max $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$	Min $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$	Max $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$	Min $\Delta L^{P}_{max}/\Delta L^{O}_{max}$	Max $\Delta L^{P}_{max}/\Delta L^{O}_{max}$
Braced frames	V-braces	0.757	1.957	0.506	2.729	0.579	1.643
	X-braces	0.715	1.488	0.772	1.252	0.772	1.237

Table 7
Minimum and maximum values of the ratios $\chi^{P}_{max}/\chi^{O}_{max}$, $\Delta L^{P}_{max}/\Delta L^{O}_{max}$ (artificial accelerograms).

Frame	Members	Positive values		Negative values		Absolute values	
		Min $\chi^{P,+}_{max}/\chi^{O,+}_{max}$	Max $\chi^{P,+}_{max}/\chi^{O,+}_{max}$	Min $\chi^{P,-}_{max}/\chi^{O,-}_{max}$	Max $\chi^{P,-}_{max}/\chi^{O,-}_{max}$	Min $\chi^{P}_{max}/\chi^{O}_{max}$	Max $\chi^{P}_{max}/\chi^{O}_{max}$
r.c. frame	Beams	0.874	1.091	0.880	1.085	0.880	1.057
	Columns	0.926	1.128	0.871	1.062	0.905	1.128
Steel MRF	Beams	0.830	1.611	0.861	1.145	0.861	1.145
	Columns	0.835	1.176	0.872	1.133	0.879	1.133
		Min $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$	Max $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$	Min $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$	Max $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$	Min $\Delta L^{P}_{max}/\Delta L^{O}_{max}$	Max $\Delta L^{P}_{max}/\Delta L^{O}_{max}$
Braced frames	V-braces	0.673	1.470	0.260	3.233	0.277	2.283
	X-braces	0.840	1.462	0.875	1.386	0.850	1.346

Table 8
Minimum and maximum values of the ratio $\Delta u^{P}_{max}/\Delta u^{O}_{max}$.

Frame	Natural accelerograms					
	Min $\Delta u^{P,+}_{max}/\Delta u^{O,+}_{max}$	Max $\Delta u^{P,+}_{max}/\Delta u^{O,+}_{max}$	Min $\Delta u^{P,-}_{max}/\Delta u^{O,-}_{max}$	Max $\Delta u^{P,-}_{max}/\Delta u^{O,-}_{max}$	Min $\Delta u^{P}_{max}/\Delta u^{O}_{max}$	Max $\Delta u^{P}_{max}/\Delta u^{O}_{max}$
r.c. frame	0.917	1.067	0.947	1.045	0.917	1.067
Steel MRF	0.958	1.101	0.993	1.203	0.939	1.086
Braced frame (V-CBF)	0.848	1.737	0.757	1.418	0.757	1.264
Braced frame (X-CBF)	0.785	1.180	0.909	1.241	0.867	1.104
	Artificial accelerograms					
	Min $\Delta u^{P,+}_{max}/\Delta u^{O,+}_{max}$	Max $\Delta u^{P,+}_{max}/\Delta u^{O,+}_{max}$	Min $\Delta u^{P,-}_{max}/\Delta u^{O,-}_{max}$	Max $\Delta u^{P,-}_{max}/\Delta u^{O,-}_{max}$	Min $\Delta u^{P}_{max}/\Delta u^{O}_{max}$	Max $\Delta u^{P}_{max}/\Delta u^{O}_{max}$
r.c. frame	0.953	1.063	0.915	1.013	0.915	1.051
Steel MRF	0.927	1.122	0.928	1.036	0.927	1.133
Braced frame (V-CBF)	0.546	1.730	0.623	1.761	0.697	1.194
Braced frame (X-CBF)	0.883	1.247	0.886	1.278	0.883	1.120

the chevron configuration. In this case, the maximum ratio $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$ is even higher than 3.23 while the maximum ratios $\Delta L^{P}_{max}/\Delta L^{O}_{max}$ are 1.643 and 2.283 for natural and artificial accelerograms, respectively. The differences in the structural behaviour of this frame appear to be strongly amplified by the degradation of strength and stiffness that occur after brace buckling. In the case of the frame with tension only braces, the maximum ratios $\Delta L^{P,+}_{max}/\Delta L^{O,+}_{max}$ and $\Delta L^{P,-}_{max}/\Delta L^{O,-}_{max}$ are still fairly high and in the range from 1.252 to 1.488. The ratios $\Delta L^{P}_{max}/\Delta L^{O}_{max}$ range from 0.772 to 1.346 (Tables 6 and 7).

Finally, the differences between the results provided by the two models of steel in terms of storey drifts are shown in Table 8. In keeping with the results obtained for the frame with BRBs, the maximum differences in terms of storey drifts are lower than those in terms of curvature of flexural members or axial elongation/shortening of braces. Notwithstanding, the maximum differences in terms of storey drifts are still significant in the frames with concentric braces in the chevron configuration.

To investigate further the effect of the proposed model on the seismic behaviour of the structures, the average of the maximum responses (e.g. the curvatures of each cross-section of the beams of the moment resisting frame or the axial deformations of braces) is calculated over a variable number of accelerograms. In this regard Eurocode 8 stipulates that if the response is obtained from at least 7 nonlinear time-history analyses, the average of the response quantities from all of these analyses should be used as the design value of the action effect. Otherwise, the most unfavourable value of the response quantity among the analyses should be considered as the design value. In any case, the number of nonlinear time-history analyses cannot be lower than three. In this paper, to show the most unfavourable average results deriving from the numerical analyses considered, the accelerograms are first specifically ordered for each cross-section (or brace), from that causing the maximum ratio between the responses of the two models to that causing the minimum ratio. Then, the averages of the response values resulting from the two models are separately calculated over an increasing number of the ordered accelerograms and their ratio is reported in Fig. 18. The number of the accelerograms ranges from 1 to 20 in the case of the natural ground motions and from 1 to 10 in the case of the artificially generated accelerograms. To consider the maximum underestimate in the above average ratios, the previously ordered accelerograms are also considered in the reverse order. Further in Fig. 18a and b, the ratios $\chi^{P,+}_{max}/\chi^{O,+}_{max}$ and $\chi^{P,-}_{max}/\chi^{O,-}_{max}$ obtained for positive and negative curvatures of the single cross-section of the r.c. and steel moment resisting frame are plotted by two different lines. A similar procedure is used to compare the maximum elongation or shortening of braces (Fig. 18c–e). Owing to the procedure applied for the calculation of the above response parameters, the higher the number of the accelerograms, the lower the difference between unity and the values reported in the figures. In any case, if the number of accelerograms is 7 (i.e. the minimum value suggested by Eurocode 8 to consider the average values as the design action effect), the difference between the responses predicted by the two versions of the steel model can be significant in steel structures. In fact, the results produced by 7 accelerograms may differ up to 20% in

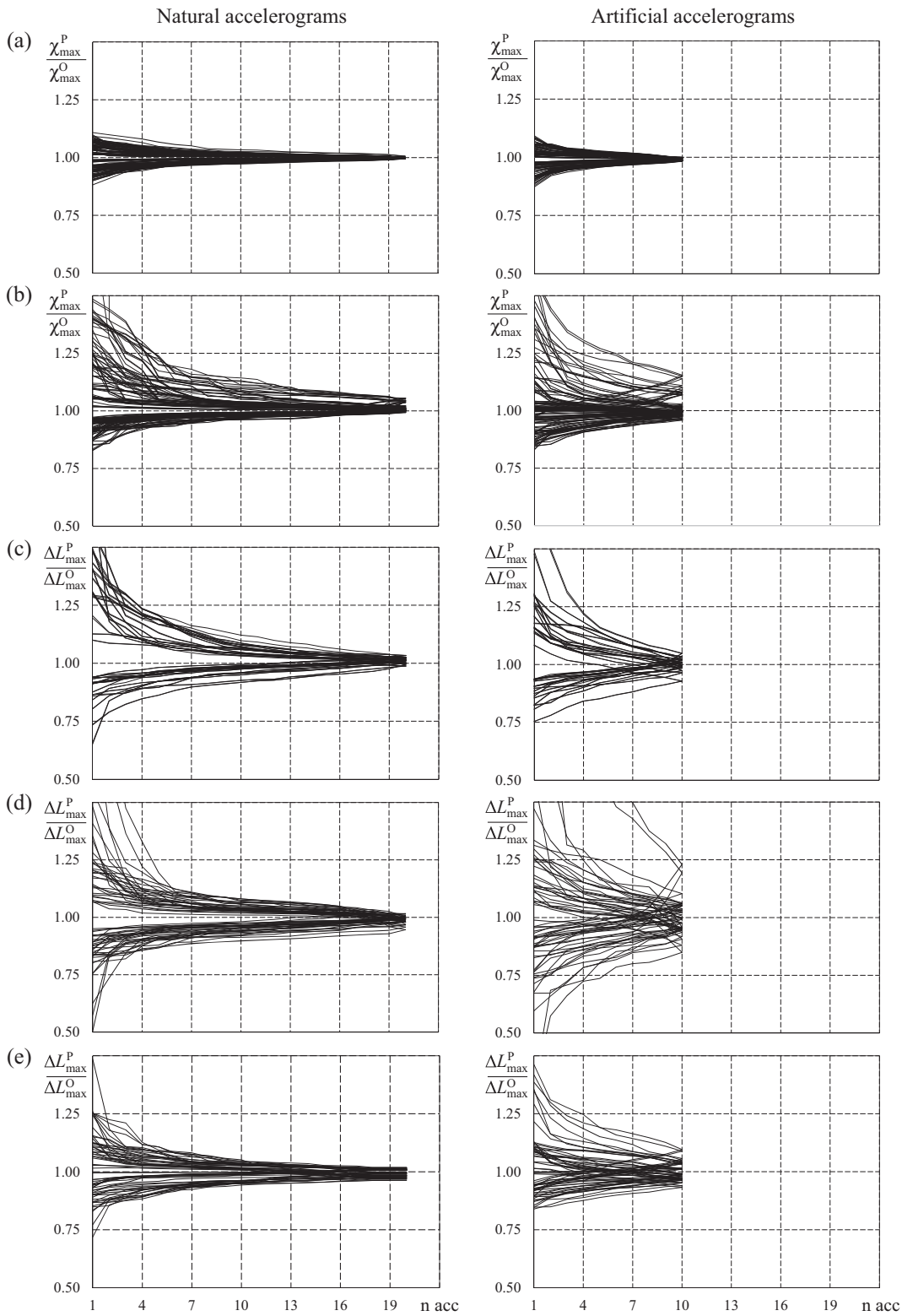

Fig. 18. Ratios of the averages of the maximum response over a variable number of accelerograms: (a) r.c. moment resisting frame, (b) steel moment resisting frame, (c) buckling restrained braced frame, (d) concentrically chevron braces frame, (e) tension only braced frame.

the steel moment resisting frame, up to 18% in the braced frames subjected to the natural accelerograms and up to 50% in the braced frames subjected to the artificially generated accelerograms. Instead, the difference between the responses predicted by the two versions of the steel model is negligible in the r.c. structure. Finally, note that the highest and lowest values shown in Fig. 18 (number of accelerograms equal to one) are those reported in Tables 5–7.

7. Conclusion

In this paper, a modification of the model proposed by Menegotto and Pinto [1] for steel is suggested to improve the response in case of reloading after partial unloading. The modified model retains the simplicity of the original formulation and is adequate for use in structural analysis. The validity of the new formulation is first demonstrated by comparison of the response of the current and new models on some reinforced concrete and steel structures. The prediction obtained by the proposed model is also compared to that obtained by the uniaxial material model recently proposed by Zsarnóczay and Budaházy to improve the response of the Menegotto and Pinto model in case of reloading after partial unloading.

The main conclusions of the research study are the following:

- The effects of the flaw present in the formulation proposed by Menegotto and Pinto appear to be negligible for reinforced concrete structures.
- The original model proposed by Menegotto and Pinto can cause significant errors in the response of steel moment resistant structures and braced structures.
- The errors tend to decrease significantly when the results are averaged over a number of accelerograms.
- The proposed model appears to be more effective than that proposed by Zsarnóczay and Budaházy in simulating the cyclic response in the occurrence of reloading after partial unloading. Further, the proposed model requires a lower calculus memory
- The computational time required by the proposed model is only slightly longer than that required by the original Menegotto and Pinto model.

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